

Week 2 · Class 1: Practice Exercises

Introduction to R and Data Frames

2025-12-31

1 Segment A — Introduction to R and Data Frames

2 Non-AI Exercises

2.1 1. Basic R Operations and Variables

2.1.1 1.1 Match Data Types

What type of data would each of these variables contain?

- a) `vote_count <- 538`
- b) `party_name <- "Democratic"`
- c) `is_incumbent <- TRUE`
- d) `approval_rating <- "45%"`

Options: numeric, character, logical

Answers: a = _____, b = _____, c = _____, d = _____

2.1.2 1.2 Fill in the Blanks: Vector Creation

Complete this code to create a vector of battleground states:

```
battleground_states <- __("Texas", _____, "Pennsylvania", _____, _____)
```

Missing states: Florida, Ohio, Michigan

2.1.3 1.3 Code Detective: Vector Operations

What does this code do? Explain each line:

```
poll_results <- c(48, 52, 45, 51, 49)
average_result <- mean(poll_results)
max_result <- max(poll_results)
results_above_50 <- poll_results > 50
```

Line 1: _____ Line 2: _____ Line 3: _____ Line 4: _____

2.2 2. Understanding Data Frames

2.2.1 2.1 Data Frame Structure

When you see this output from `glimpse()`:

```
Rows: 100
Columns: 5
$ candidate    <chr> "Smith", "Jones", "Brown"...
$ party        <chr> "Dem", "Rep", "Dem"...
$ votes        <dbl> 15234, 16789, 14556...
$ district     <int> 1, 1, 2...
$ incumbent    <lgl> TRUE, FALSE, FALSE...
```

Answer these questions: - How many observations are in this dataset? _____ - How many variables are in this dataset? _____ - What type of data is stored in the `votes` column? _____ - Which column contains TRUE/FALSE values? _____

2.2.2 2.2 True or False: Data Frames

Mark each statement as True (T) or False (F):

_____ Data frames can only contain numeric data _____ Each column in a data frame must have the same length _____ Rows represent observations, columns represent variables _____ You can mix different data types in the same column _____ `glimpse()` shows you the structure and first few values

2.3 3. Data Types and Functions

2.3.1 3.1 Matching Data Types

Match each data type abbreviation with its meaning:

Abbreviations: a) <dbl> b) <chr> c) <int> d) <lgl> e) <date>

Meanings: 1. Whole numbers only 2. TRUE/FALSE values 3. Text data 4. Decimal numbers 5. Calendar dates

Matches: a = _____, b = _____, c = _____, d = _____, e = _____

2.3.2 3.2 Multiple Choice: Functions

Which function would you use to:

1. See the first 6 rows of data: _____
2. Get column names: _____
3. Get summary statistics: _____
4. See data structure and types: _____
5. Round to 2 decimal places: _____

Options: a) head(), b) names(), c) summary(), d) glimpse(), e) round(x, 2)

2.4 4. Error Diagnosis

2.4.1 4.1 Spot the Error

What's wrong with each code snippet?

```
# Error 1
Mean(poll_results)

# Error 2
numbers <- c("1", "2", "3")
mean(numbers)

# Error 3
read_csv("data.csv")
```

Error 1 problem: _____ Error 2 problem: _____ Error 3 problem: _____

2.4.2 4.2 Multiple Choice: Debugging

In the code below, why does line 2 produce an error?

```
electoral_votes <- c(20, 16, 10, 11, 16, 6)
average_votes = mean(Electoral_Votes)
```

- a) The mean() function doesn't exist
- b) Variable names are case-sensitive (electoral_votes Electoral_Votes)
- c) You can't calculate mean of a vector
- d) The assignment operator should be <-

Answer: _____

3 AI Exercises

Tips for Working with Claude:

- Ask for **R code using only tidyverse** (no other packages)
- Request **simple, focused answers** to your specific question—not complex analyses
- Ask Claude to **explain what the code is doing** since you're learning
- Avoid asking for visualizations or plots in these exercises
- Include the output of `glimpse()` in your prompt so Claude knows your variable names

Example prompt: “Using tidyverse in R, calculate the mean years_served by party. Keep the code simple and explain what each line does. Here is what my data looks like: [paste glimpse output]”

3.1 5. Working with Congressional Data

Dataset: data/congressional_leaders.csv

Description: Dataset containing information about current Congressional leadership.

Variables: - `member_id`: Unique identifier (int) - `name`: Leader's name (chr) - `party`: Political party (chr) - `chamber`: House or Senate (chr) - `state`: State represented (chr) - `years_served`: Years in Congress (int) - `age`: Current age (int) - `committee_chair`: Whether they chair a committee (lgl) - `leadership_position`: Their leadership role (chr)

3.1.1 5.1 Loading and Exploring Data

```
# Load the dataset
library(tidyverse)
leaders <- read_csv("data/congressional_leaders.csv")
```

Rows: 4 Columns: 9

-- Column specification -----

Delimiter: ","

chr (5): name, party, chamber, state, leadership_position

dbl (3): member_id, years_served, age

lgl (1): committee_chair

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Explore the data
glimpse(leaders)
```

Rows: 4

Columns: 9

```
$ member_id      <dbl> 1, 2, 3, 4
$ name           <chr> "Pelosi", "Schumer", "McConnell", "McCarthy"
$ party          <chr> "Democratic", "Democratic", "Republican", "Republi~
$ chamber        <chr> "House", "Senate", "Senate", "House"
$ state          <chr> "CA", "NY", "KY", "CA"
$ years_served   <dbl> 36, 24, 38, 16
$ age            <dbl> 82, 72, 80, 58
$ committee_chair <lgl> FALSE, FALSE, FALSE, FALSE
$ leadership_position <chr> "Speaker", "Majority Leader", "Minority Leader", "~
```

```
head(leaders)
```

A tibble: 4 x 9

	member_id	name	party	chamber	state	years_served	age	committee_chair
	<dbl>	<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<lgl>
1	1	Pelosi	Democrat~	House	CA	36	82	FALSE
2	2	Schumer	Democrat~	Senate	NY	24	72	FALSE
3	3	McConnell	Republic~	Senate	KY	38	80	FALSE
4	4	McCarthy	Republic~	House	CA	16	58	FALSE

i 1 more variable: leadership_position <chr>

Use Claude to help you understand this Congressional leadership dataset and calculate summary statistics about years of service and age.

3.1.2 5.2 Basic Analysis

Ask Claude to help you find: - The average years of service by party - Which chamber has older members on average - How many committee chairs there are

3.1.3 5.3 Creating Summaries

Work with Claude to create a summary table showing the number of leaders by party and chamber.

3.2 6. State Voting Patterns Analysis

Dataset: data/state_voting_patterns.csv

Description: State-level voting patterns in recent presidential elections.

Variables: - state: State name (chr) - state_abbr: Two-letter state code (chr) - region: Geographic region (chr) - electoral_votes: Number of electoral votes (int) - dem_2020: Democratic vote share 2020 (dbl) - rep_2020: Republican vote share 2020 (dbl) - dem_2016: Democratic vote share 2016 (dbl) - rep_2016: Republican vote share 2016 (dbl) - swing_state: Whether considered a swing state (lgl) - population_millions: State population in millions (dbl)

3.2.1 6.1 Initial Data Exploration

```
# Load the dataset
voting <- read_csv("data/state_voting_patterns.csv")
```

Rows: 50 Columns: 10

-- Column specification -----

Delimiter: ","

chr (3): state, state_abbr, region

dbl (6): electoral_votes, population_millions, dem_2020, rep_2020, dem_2016,...

lgl (1): swing_state

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Examine the structure
glimpse(voting)
```

```
Rows: 50
Columns: 10
$ state      <chr> "Alabama", "Alaska", "Arizona", "Arkansas", "Calif~
$ state_abbr <chr> "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "F~
$ region     <chr> "South", "West", "West", "South", "West", "West", ~
$ electoral_votes <dbl> 9, 3, 11, 6, 54, 10, 7, 3, 30, 16, 4, 4, 19, 11, 6~
$ population_millions <dbl> 5.0, 0.7, 7.3, 3.0, 39.5, 5.8, 3.6, 1.0, 21.8, 10.~
$ dem_2020    <dbl> 35.8, 41.1, 51.1, 33.9, 67.5, 57.8, 60.4, 56.4, 46~
$ rep_2020    <dbl> 62.8, 57.2, 48.3, 65.2, 30.5, 40.8, 37.7, 41.2, 51~
$ dem_2016    <dbl> 34.7, 38.1, 47.5, 31.3, 63.0, 56.3, 55.5, 54.3, 45~
$ rep_2016    <dbl> 62.0, 58.7, 48.4, 64.8, 33.8, 40.7, 41.7, 41.7, 49~
$ swing_state <lgl> FALSE, FALSE, TRUE, FALSE, FALSE, FALSE, FALSE, FA~
```

```
summary(voting)
```

state	state_abbr	region	electoral_votes
Length:50	Length:50	Length:50	Min. : 3.00
Class :character	Class :character	Class :character	1st Qu.: 4.25
Mode :character	Mode :character	Mode :character	Median : 8.00
			Mean :10.70
			3rd Qu.:11.75
			Max. :54.00
population_millions	dem_2020	rep_2020	dem_2016
Min. : 0.600	Min. :31.10	Min. :30.00	Min. :26.40
1st Qu.: 1.925	1st Qu.:40.35	1st Qu.:42.42	1st Qu.:37.42
Median : 4.550	Median :49.65	Median :48.95	Median :46.55
Mean : 6.610	Mean :48.41	Mean :50.06	Mean :45.67
3rd Qu.: 7.600	3rd Qu.:56.15	3rd Qu.:58.92	3rd Qu.:53.77
Max. :39.500	Max. :69.30	Max. :67.40	Max. :67.90
rep_2016	swing_state		
Min. :29.10	Mode :logical		
1st Qu.:43.42	FALSE:41		
Median :50.40	TRUE :9		
Mean :50.88			
3rd Qu.:60.27			
Max. :70.10			

Work with Claude to explore state voting patterns. What questions would you ask to understand partisan geography?

3.2.2 6.2 Calculating Vote Margins

Ask Claude to help you: - Calculate the margin of victory in 2020 for each state (Republican % - Democratic %) - Find which states had the closest elections (smallest margins)

3.2.3 6.3 Swing State Analysis

Work with Claude to analyze differences between swing states and non-swing states in terms of population and electoral votes.

3.3 7. Public Opinion Tracking

Dataset: data/presidential_approval.csv

Description: Weekly presidential approval ratings over time.

Variables: - **week_id:** Week identifier (int) - **date:** Date of poll (date) - **president:** President's name (chr) - **party:** President's party (chr) - **approve:** Approval percentage (dbl) - **disapprove:** Disapproval percentage (dbl) - **unsure:** Unsure percentage (dbl) - **sample_size:** Number of respondents (int) - **pollster:** Polling organization (chr) - **economic_index:** Economic conditions index (dbl)

3.3.1 7.1 Loading and Understanding the Data

```
# Load the dataset
approval <- read_csv("data/presidential_approval.csv")
```

Rows: 528 Columns: 10

-- Column specification -----

Delimiter: ","

chr (3): president, party, pollster

dbl (6): week_id, approve, disapprove, unsure, sample_size, economic_index

date (1): date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Check the data
glimpse(approval)
```



```

Rows: 528
Columns: 10
$ week_id      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, ~
$ date         <date> 1980-01-15, 1980-02-15, 1980-03-15, 1980-04-15, 1980-0~
$ president    <chr> "Obama", "Obama", "Obama", "Obama", "Obama", "Obama", "~
$ party        <chr> "Democrat", "Democrat", "Democrat", "Democrat", "Democr~
$ approve       <dbl> 58.2, 55.8, 39.4, 40.1, 43.2, 31.8, 33.1, 32.8, 37.5, 3~
$ disapprove    <dbl> 34.3, 36.6, 56.9, 52.9, 50.9, 63.1, 60.5, 64.4, 56.6, 5~
$ unsure        <dbl> 7.5, 7.6, 3.7, 7.0, 5.9, 5.1, 6.4, 2.8, 5.9, 6.2, 4.7, ~
$ sample_size   <dbl> 1436, 1882, 1636, 1481, 1501, 1985, 886, 1027, 1997, 19~
$ pollster      <chr> "Fox News", "YouGov", "Pew", "Gallup", "Fox News", "Mor~
$ economic_index <dbl> 51.3, 51.4, 38.3, 45.2, 47.7, 46.1, 40.1, 33.1, 36.3, 4~

```

```
names(approval)
```

```

[1] "week_id"      "date"          "president"     "party"
[5] "approve"      "disapprove"    "unsure"        "sample_size"
[9] "pollster"     "economic_index"

```

This dataset tracks presidential approval over time. Ask Claude to help you understand the structure and calculate key metrics.

3.3.2 7.2 Trend Analysis

Ask Claude to help you: - Calculate the average approval rating by president - Find the highest and lowest approval ratings for each president - Identify any patterns in the “unsure” responses

3.3.3 7.3 Data Quality Check

Work with Claude to: - Verify that approve + disapprove + unsure = 100 for each row - Identify any data quality issues - Create a summary of findings

3.4 8. Campaign Finance Exploration

Dataset: data/campaign.csv

Description: Campaign contribution data for federal candidates.

Variables: - contribution_id: Unique identifier (int) - candidate_name: Recipient candidate (chr) - candidate_party: Party affiliation (chr) - office: Office sought (chr) - state:

State (chr) - amount: Contribution amount in dollars (dbl) - date: Date of contribution (date)
 - contributor_type: Individual, PAC, etc. (chr) - contributor_state: Contributor's state
 (chr) - is_corporate: Whether from corporate PAC (lgl)

3.4.1 8.1 Initial Data Load

```
# Load the dataset
campaign <- read_csv("data/campaign.csv")
```

Rows: 2874 Columns: 50

-- Column specification -----

Delimiter: ","

chr (11): candidate_name, candidate_party, office, state, contributor_type,...

dbl (37): contribution_id, amount, cycle, cfscore, male.winner, running.var...

lgl (1): is_corporate

date (1): date

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Explore campaign finance data
glimpse(campaign)
```

Rows: 2,874

Columns: 50

```
$ contribution_id      <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, ~
$ candidate_name       <chr> "LAUGHLIN, SUSAN", "OPAKE, MICHAEL A", "ROB~
$ candidate_party      <chr> "Democrat", "Democrat", "Republican", "Demo~
$ office               <chr> "state:lower", "state:upper", "state:upper"~
$ state                <chr> "PA", "PA", "PA", "PA", "PA", "PA", "PA", "~
$ amount               <dbl> 40375, 228235, 206336, 560965, 14940, 55563~
$ date                 <date> 1998-01-14, 2004-05-13, 1998-01-13, 2004-0~
$ contributor_type     <chr> "Party Committee", "Party Committee", "PAC"~
$ contributor_state    <chr> "VA", "UT", "OK", "MI", "CA", "OK", "NY", "~
$ is_corporate         <lgl> FALSE, TRUE, TRUE, TRUE, TRUE, FALSE, TRUE,~
$ bonica.rid           <chr> "cand100708", "cand101663", "cand101666", "~
$ cycle                <dbl> 1998, 2004, 1998, 2004, 2008, 2002, 1998, 1~
$ election              <chr> "PA1998", "PA2004", "PA1998", "PA2004", "PA~
$ Cand.ID              <chr> "PA105027", "PA1080", "PA1083", "PA1086", "~
$ District             <chr> "PA-16", "PA-11", "PA-50", "PA-7", "PA-109"~
```

```

$ cfscore <dbl> -0.10653670, 0.11879573, 0.60734081, -0.390~
$ male.winner <dbl> 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1~
$ running.variable <dbl> -15.259799, 20.875214, 18.254158, 33.552872~
$ female.margin <dbl> 65.25980, 29.12479, 31.74584, 16.44713, 38.~
$ female.pctile <dbl> 65.3, 29.1, 31.7, 16.4, 38.3, 30.7, 58.6, 8~
$ prev.elect <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
$ democrat <dbl> 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0, 1~
$ total.votes <dbl> 16.609, 98.696, 58.625, 109.320, 25.872, 23~
$ professional <dbl> 0.283, 0.283, 0.283, 0.283, 0.283, 0.283, 0~
$ pct.female.chamber <dbl> 0.1594684, 0.1594203, 0.2307692, 0.1594203,~
$ gop.gov <dbl> 1, 0, 1, 0, 0, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1~
$ house <dbl> 1, 0, 0, 0, 1, 1, 1, 1, 0, 0, 1, 0, 1, 1, 1~
$ state.pres.dem <dbl> 0.4917, 0.5092, 0.4917, 0.5092, 0.5447, 0.5~
$ num.cands <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2~
$ won.again <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
$ pct.dem.last <dbl> NA, 94.89754, NA, 100.00000, 41.50056, 0.00~
$ female.cand.last <dbl> NA, 0, NA, 0, 0, 0, NA, NA, NA, 1, 0, NA, N~
$ total.dist.last <dbl> NA, 108629, NA, 87128, 49161, 12468, NA, NA~
$ female.inc <dbl> 1, 0, 0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0~
$ total.raised.candidate <dbl> 40375, 228235, 206336, 560965, 14940, 55563~
$ log.total.raised.candidate <dbl> 10.605966, 12.338131, 12.237261, 13.237414,~
$ individual.money <dbl> 3925, 108032, 69259, 128380, 1510, 6130, 12~
$ log.individual.money <dbl> 8.275376, 11.590192, 11.145623, 11.762758, ~
$ pac.money <dbl> 36000, 95050, 83200, 418700, 12100, 15375, ~
$ log.pac.money <dbl> 10.491302, 11.462169, 11.329015, 12.944912,~
$ party.money <dbl> 250, 17436, 22644, 0, 500, 32458, 200, 0, 1~
$ log.party.money <dbl> 5.525453, 9.766350, 10.027694, 0.000000, 6.~
$ male.money <dbl> 2625.00, 102631.60, 61289.00, 102905.00, 13~
$ log.male.money <dbl> 7.873217, 11.538911, 11.023372, 11.541571, ~
$ female.money <dbl> 1300.00, 4800.00, 3855.00, 26850.00, 200.00~
$ log.female.money <dbl> 7.170888, 8.476580, 8.257386, 10.198058, 5.~
$ share.district.total <dbl> 1.0000000, 0.7638080, 0.9615315, 0.9399154,~
$ district.total <dbl> 40375, 298812, 214591, 596825, 23172, 55563~
$ log.district.total <dbl> 10.60597, 12.60757, 12.27649, 13.29938, 10.~
$ cand.gender <chr> "F", "M", "M", "M", "M", "M", "F", "F", "F"~

```

Campaign finance data can reveal patterns in political support. Work with Claude to explore contribution patterns.

3.4.2 8.2 Contribution Patterns

Ask Claude to help you analyze: - Average contribution size by contributor type - Total contributions by party - Which states contribute the most

3.4.3 8.3 Creating a Summary Report

Work with Claude to create a brief summary report about campaign finance patterns, including key statistics and any interesting findings.